

The implementation of an algorithm to calculate thermodynamic equilibria for multi-component systems with non-ideal phases in a free software



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ABSTRACT

Thermodynamics is a central part of materials science. Thermodynamic models provide a unique method to combine experimental data and results from first-principles calculations in databases. Equilibrium calculations based on these databases are essential to provide values of many different thermodynamic properties for software tools for simulating materials processes and to predict their final properties. It is important for such work to have a free software with access to the source code.

In this paper, a well-established algorithm to calculate thermodynamic equilibria for multi-component systems using different kinds of conditions for non-ideal solution phases with different models is explained in detail, including changes of the set of stable phases during iterations. A new definition of the molar Gibbs energy and a powerful method to handle composition variables and phase amounts are presented. The technique to calculate derivatives of state variables based on the first and second derivatives of the Gibbs energy is explained.

The algorithm can also be used to calculate properties outside the equilibrium state as required for the simulation of phase transitions.

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1. Introduction

Computational thermodynamics is an important part of materials science and the practical use of software and databases is well described by Saunders and Miodownik [1] and Kaufman [2] and models, assessments techniques and database development are discussed in Lukas et al. [3].

Several algorithms for equilibrium calculations are also available, and recently one was presented by Piro et al. [4]. The algorithm presented here was first derived in slightly different forms by Eriksson and Rosen [5], Hillert [6], in the thesis by Jansson [7] and used in a paper by Lukas et al. [8] and recently by Shobu and Taburu [9]. The paper by Hillert [6] gives a detailed theoretical base but all papers are very terse and difficult to understand. In this paper the algorithm and its implementation are explained in detail because many users of thermodynamics frequently make mistakes with partial derivatives, constraints and conditions. We make a clear separation of global variables and those which depend on the model. Additional features like the global

minimization technique and a powerful method to use the result of an equilibrium calculation to calculate properties like the real heat capacity are also explained.

The implementation is done as part of the Open Calphad (OC) initiative [10] to provide a free software for thermodynamic calculations. This open source software has a GNU license and is available on the web [11,12] and the intention is to integrate it in several different software for example to use results from first-principles calculations [13–15], in assessments for databases [16] as well as for simulations of phase transformations and microstructures including kinetic data [17,18] and in industrial applications [19–21]. The principal goals are that the OC software should be stable, accurate and independently maintainable by those interested. It is prepared for parallelization and can calculate an equilibrium as fast as comparable commercial software. Procedures for the calculation of property and phase diagrams based on the OC equilibrium software are available.

2. The thermodynamic model

Phases in a thermodynamic system can be very different, examples are gas, liquid, amorphous and many different crystalline

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structures. In a closed system at constant T and P the equilibrium is given by the minimum in the Gibbs energy, G . This can be expressed as a sum of the molar Gibbs energies for the stable phases, α :

$$G = \sum_{\alpha} N^{\alpha} G_M^{\alpha}(T, P, Y) \quad (1)$$

where N^{α} is the number of moles of formula units of the phase α and G_M^{α} is the Gibbs energy per mole formula unit of α . The Gibbs energy of each phase can be modeled differently but it depends on T , P and its constitution, Y . Note that the definition of the molar Gibbs energy here is different from the normal

$$G_m = \frac{G}{N} \quad (2)$$

where N is the total number of moles of components of the system. In this paper G_M^{α} will be used according to the definition in Eq. (1), using the formula unit explained in Section 2.2. To relate the Gibbs energy to the formula unit of the phase rather than the amount of components in the phase significantly simplifies the equilibrium calculations as well as the modeling of the phases.

For other external conditions like fixed volume, chemical potential, etc. we can add constraints to obtain the appropriate Lagrange functions to minimize. Internal constraints, that the sum of constituent fractions on each sublattice is unity and the net charge for each phase is zero, can also be handled by Lagrange multipliers.

The models used by the minimizer are implemented in a separate free software package, the General Thermodynamic Package (GTP) and this, as well as its documentation, is available on the website [11]. The explanations in this paper will be based on models which are part of the Compound Energy Formalism (CEF) [3,22,23]. The CEF formalism includes almost all practically useful models to describe gases, substitutional regular solutions, interstitial solutions, and sublattice models, magnetism and defect models, etc. as explained in the book by Lukas et al. [3]. For the ionic liquid model [3,24], which is also implemented in GTP, some special features of the minimization algorithm have been added as described in the full documentation of the code [11]. A well-defined software interface between the model package and the minimizer to simplify the implementation of new models is available.

2.1. Elements, species, constituents and components

The meaning of some terms used in this paper is given here:

- Elements are those on the periodic chart or fictitious denoted by symbols like A, B, etc.
- Species is a molecular-like aggregate of the elements with fixed stoichiometry and possibly with a charge. The simplest species is an element. A special species called vacancy and denoted “Va” is used to denote empty sites in a phase. There is no mass-balance equation for the vacancy but at equilibrium the chemical potential of the vacancy is zero.
- The constituents of a phase are a subset of the species. A phase may have the same species as constituent in several sublattices. A phase with fixed composition may have a single species as constituent or a single constituent in each sublattice.
- Components are usually the same as elements but for convenience one may redefine them to any orthogonal set of species, like using the components CaO, SiO₂ and O rather than Ca, Si and O. In the text below “element” will be used as equivalent to component.

The separation of composition variables for the modeling, the constituent fraction, and those controlled externally, like the amount or mole or mass fraction of components, is an essential feature of the implementation of the minimizer and the models.

2.2. The formula unit of a phase

For a phase modeled with CEF we can have sublattices and the ratios of the sites in each sublattice are denoted a_s , where s is the sublattice. For example the σ phase has 5 sublattices with a_s equal to 2, 4, 8, 8 and 8 sites. Site ratios can be integers or fractions but the recommendation is to use the smallest integer values.

The constituent fractions for a phase α are denoted y_{is}^{α} where i is the constituent and s the sublattice. When there is no possible confusion the phase superscripts and sublattice subscripts will be omitted and a summation over the constituents i will mean for all constituents in all sublattices of a phase.

We will from now on use indices A, B, etc. to denote elements (or components) and indices i , j , etc. to denote constituents of a phase. In some cases the constituents of a phase are the same as the elements. To avoid confusion between the two similar terms “component” and “constituent”, the term “element” will be used in the text below also to mean component.

The number of moles of an element A per mole formula unit of the phase, M_A^{α} is calculated as

$$M_A^{\alpha} = \sum_s a_s^{\alpha} \sum_i b_{Ai} y_{is}^{\alpha} \quad (3)$$

where b_{Ai} is the stoichiometric coefficient of element A in constituent i and y_{is}^{α} the fraction of constituent i in sublattice s of the phase α . a_s^{α} is the number of sites for sublattice s in the phase α . The sum of constituent fractions on each sublattice is unity:

$$\sum_i y_{is}^{\alpha} = 1 \quad (4)$$

The mole fraction of element A in the α phase is

$$x_A^{\alpha} = \frac{M_A^{\alpha}}{\sum_B M_B^{\alpha}} = \frac{M_A^{\alpha}}{M^{\alpha}} \quad (5)$$

Provided we have only the elements as constituents we can use

$$M^{\alpha} = \sum_s a_s^{\alpha} (1 - y_{Va,s}^{\alpha}) \quad (6)$$

where $y_{Va,s}^{\alpha}$ is the fraction of vacancies on sublattice s .

The total number of moles of an element A, N_A , in a system with several phases is

$$N_A = \sum_{\alpha} N^{\alpha} M_A^{\alpha} \quad (7)$$

where, as before, N^{α} is the number of moles of formula unit of the phase α . The total number of moles of elements, N , and the overall mole fraction of element A, x_A , in the system are

$$N = \sum_A N_A = \sum_A \sum_{\alpha} N^{\alpha} M_A^{\alpha} \quad (8)$$

$$x_A = \frac{N_A}{N} \quad (9)$$

We have to be careful to distinguish between N , N , M , x , y with and without superscripts and indices.

2.3. Differentials

The differential of M_A^{α} is

$$dM_A^{\alpha} = \sum_s \sum_i \left(\frac{\partial M_A^{\alpha}}{\partial y_{is}^{\alpha}} \right) dy_{is}^{\alpha} \quad (10)$$

$$\left(\frac{\partial M_A^{\alpha}}{\partial y_{is}^{\alpha}} \right)_{y_{jt \neq is}} = a_s^{\alpha} b_{Ai} \quad (11)$$

$$\sum_i dy_{is}^{\alpha} = 0 \quad (12)$$

Most of the derivatives $\left(\frac{\partial M_A^x}{\partial y_{is}^x}\right)_{y_{it \neq is}}$ will be zero as M_A often depends on a single or just a few y_{is} . But we can have molecules with several elements as constituents and also vacancies in a sublattice and this means the number of moles of elements per formula unit of the phase is often variable. For the ionic liquid model the situation is different as the number of sites for cations and anions depends on the average charge on the other sublattice but it does not affect the general procedure for the minimization of the Gibbs energy described here.

2.4. Examples

Understanding the meaning of the term “formula unit” is essential to the algorithm so a few examples will be given. In each example we assume the system consists of a single phase and ignore phase labels.

2.4.1. A gas phase with H and O

In a gas with the molecules (H_2 , O_2 , H_2O) the moles of the elements H and O and the total number of moles in a formula unit, M , are:

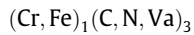
$$\begin{aligned} M_H &= 2y_{H_2} + 2y_{H_2O} \\ M_O &= 2y_{O_2} + y_{H_2O} \\ M &= 2y_{H_2} + 2y_{O_2} + 3y_{H_2O} \end{aligned} \quad (13)$$

The number of moles of elements per formula unit of this gas can thus vary between 2 and 3. The mole fractions in the gas are:

$$\begin{aligned} x_H &= \frac{M_H}{M} = \frac{2y_{H_2} + 2y_{H_2O}}{2y_{H_2} + 2y_{O_2} + 3y_{H_2O}} \\ x_O &= \frac{M_O}{M} = \frac{2y_{O_2} + y_{H_2O}}{2y_{H_2} + 2y_{O_2} + 3y_{H_2O}} \end{aligned} \quad (14)$$

2.4.2. A crystalline phase with substitutional and interstitial constituents

An interstitial solution of C and N in the bcc phase in a steel with Cr is modeled.



where Va denotes a vacancy or vacant site. The moles of the elements per formula unit are:

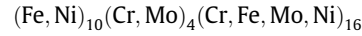
$$\begin{aligned} M_C &= 3y_{C,2} \\ M_{Cr} &= y_{Cr,1} \\ M_{Fe} &= y_{Fe,1} \\ M_N &= 3y_{N,2} \\ M &= 1 + 3y_{C,2} + 3y_{N,2} = 4 - 3y_{Va,2} \end{aligned} \quad (15)$$

The number of moles of elements per formula unit can thus vary between 1 and 4. The mole fractions in the bcc phase are:

$$\begin{aligned} x_C &= \frac{3y_{C,2}}{1 + 3y_{C,2} + 3y_{N,2}} \\ x_{Cr} &= \frac{y_{Cr,1}}{1 + 3y_{C,2} + 3y_{N,2}} \\ x_{Fe} &= \frac{y_{Fe,1}}{1 + 3y_{C,2} + 3y_{N,2}} \\ x_N &= \frac{3y_{N,2}}{1 + 3y_{C,2} + 3y_{N,2}} \end{aligned} \quad (16)$$

2.4.3. A crystalline phase with long range ordering

Finally for a σ phase in the Cr, Fe, Mo and Ni system with a simplified model using only 3 sublattices and with restricted solubility we have:



the moles of the elements per formula unit are

$$\begin{aligned} M_{Cr} &= 4y_{Cr,2} + 16y_{Cr,3} \\ M_{Fe} &= 10y_{Fe,1} + 16y_{Fe,3} \\ M_{Mo} &= 4y_{Mo,2} + 16y_{Mo,3} \\ M_{Ni} &= 10y_{Ni,1} + 16y_{Ni,3} \\ M &= 30 \end{aligned} \quad (17)$$

The number of moles of elements per formula unit is constant and equal to 30. The mole fractions in the σ phase are:

$$\begin{aligned} x_{Cr} &= \frac{4y_{Cr,2} + 16y_{Cr,3}}{30} \\ x_{Fe} &= \frac{10y_{Fe,1} + 16y_{Fe,3}}{30} \\ x_{Mo} &= \frac{4y_{Mo,2} + 16y_{Mo,3}}{30} \\ x_{Ni} &= \frac{10y_{Ni,1} + 16y_{Ni,3}}{30} \end{aligned} \quad (18)$$

2.5. End-members of models

The *end-member* is an important concept when modeling with CEF. For a crystalline phase it specifies one constituent in each sublattice of the phase. A phase with no variation in composition consists of one single end-member and for the gas phase each species is an end-member. The gas phase species can also include ions. In the bcc phase example above we have 6 end-members which can be denoted: (Cr:C), (Cr:N), (Cr:Va), (Fe:C), (Fe:N) and (Fe:Va). In the σ phase example there are 16 end-members, for example (Fe:Cr:Cr), (Fe:Cr:Fe), (Fe:Cr:Mo), (Fe:Cr:Ni), etc. Note that in most cases end-members represent compounds with several elements.

3. The Gibbs energy

The Gibbs energy, G , is an extensive property and can be formulated in many different ways. One well-known formula relates the Gibbs energy to the chemical potentials, μ_A , and the number of moles of the elements N_A :

$$G = \sum_A N_A \mu_A \quad (19)$$

The chemical potentials can be obtained from $G = G(T, P, N_A, \dots)$ as:

$$\mu_A = \left(\frac{\partial G}{\partial N_A} \right)_{T, P, N_{BA}} \quad (20)$$

If a phase has a fixed composition it has only one Gibbs energy as function of T and P for this composition and we cannot calculate the individual chemical potentials of the elements for this phase alone. But if we can vary the fraction of constituents in one or more sublattices it is possible to calculate some relations between chemical potentials as will be discussed in Section 3.2.

We will emphasize on a new definition of the molar Gibbs energy for each phase, using Eq. (3), giving the molar Gibbs energy for a formula unit of a phase α , G_M^α , and per mole of components of the phase, G_m^α :

$$G_M^\alpha = \sum_A M_A^\alpha \mu_A \quad (21)$$

$$G_m^\alpha = \sum_A x_A^\alpha \mu_A \quad (22)$$

It is important to know what kind of “molar” quantity we use. The reason for introducing the molar Gibbs energy per formula unit is because it is simpler and helps avoiding mistakes when we derive the equations for finding the equilibrium, especially when we have phases modeled with vacancies and molecules.

3.1. The differential of the Gibbs energy

The differential of the Gibbs energy is

$$dG = -SdT + VdP + \sum_A \mu_A dN_A \quad (23)$$

where S is the entropy and V the volume of the system. At constant T and P in a closed system we have $dG = 0$ at equilibrium. If we have several stable phases

$$dG = \sum_{\alpha} (\aleph^{\alpha} dG_M^{\alpha} + G_M^{\alpha} d\aleph^{\alpha}) \quad (24)$$

where dG_M^{α} for each phase using the molar Gibbs energy per formula unit can be expressed as differentials of the variables T , P and y_{is}^{α} from Eq. (10):

$$\begin{aligned} dG_M^{\alpha} &= -S_M^{\alpha} dT + V_M^{\alpha} dP + \sum_A \mu_A dM_A^{\alpha} \\ &= -S_M^{\alpha} dT + V_M^{\alpha} dP + \sum_A \mu_A \sum_s \sum_i \left(\frac{\partial M_A^{\alpha}}{\partial y_{is}^{\alpha}} \right)_{T,P,y_{jt \neq is}} dy_{is}^{\alpha} \end{aligned} \quad (25)$$

where all fractions y_{jt}^{α} except y_{is}^{α} are kept constant in the derivative. It is important to note that the constituent variables y_{is}^{α} in Eq. (25) are not independent whereas we in Eq. (23) have independent amounts of the components N_A . The consequences of this are shown in the next section.

3.2. The partial Gibbs energy for an end-member

For a phase α with sublattices it is not always possible to directly calculate the chemical potentials for the elements but we can always calculate the partial Gibbs energies of the end-members, I , of the phase. Since an end-member specifies one constituent in each sublattice, and taking into account that the constituent fractions are not independent, its chemical potential is:

$$G_I^{\alpha} = G_M^{\alpha} + \sum_s \left(\frac{\partial G_M^{\alpha}}{\partial y_{is \in I}} \right)_{T,P,y_{jt \neq is}} - \sum_s \sum_j y_{js} \left(\frac{\partial G_M^{\alpha}}{\partial y_{js}} \right)_{T,P,y_{kt \neq js}} \quad (26)$$

where the first summation is for constituents i , with fraction y_{is} , specified by end-member I for each sublattice s . The second double summation is for all constituents. Eq. (26) is not well known but was first derived by Sundman and Ågren [22] and recently by Rogal et al. [25]. For a phase with fixed composition $G_I^{\alpha} = G_M^{\alpha}$.

The model for the bcc phase $(\text{Fe}, \text{Cr})_1(\text{C}, \text{N}, \text{Va})_3$ in Section 2.4.2 does not include pure C and N but to obtain the partial Gibbs energy for C we can use the end-members (Fe:Va) and (Fe:C) in Eq. (26):

$$G_{\text{Fe:Va}} = G_M + \left(\frac{\partial G_M}{\partial y_{\text{Fe},1}} \right)_{T,P,y_{j \neq \text{Fe}}} + \left(\frac{\partial G_M}{\partial y_{\text{Va},2}} \right)_{T,P,y_{j \neq \text{Va}}} - \sum_s \sum_j y_{js} \left(\frac{\partial G_M}{\partial y_{js}} \right)_{T,P,y_{kt \neq js}} \quad (27)$$

$$G_{\text{Fe:C}} = G_M + \left(\frac{\partial G_M}{\partial y_{\text{Fe},1}} \right)_{T,P,y_{j \neq \text{Fe}}} + \left(\frac{\partial G_M}{\partial y_{\text{C},2}} \right)_{T,P,y_{j \neq \text{C}}} - \sum_s \sum_j y_{js} \left(\frac{\partial G_M}{\partial y_{js}} \right)_{T,P,y_{kt \neq js}} \quad (28)$$

At equilibrium the partial Gibbs energies of the end-members are related to the chemical potentials of the elements as

$$\begin{aligned} G_{\text{Fe:Va}_3} &= G_{\text{Fe:Va}} = G_{\text{Fe}} = \mu_{\text{Fe}} \\ G_{\text{Fe:C}_3} &= G_{\text{Fe:C}} = G_{\text{Fe}} + 3G_{\text{C}} = \mu_{\text{Fe}} + 3\mu_{\text{C}} \end{aligned} \quad (29)$$

because the chemical potential of Va, $\mu_{\text{Va}} = 0$ at equilibrium. We can rearrange to obtain the chemical potential of C:

$$\mu_{\text{C}} = G_{\text{C}} = \frac{1}{3} (G_{\text{Fe:C}} - G_{\text{Fe:Va}}) = \frac{1}{3} \left[\left(\frac{\partial G_M}{\partial y_{\text{C},2}} \right)_{T,P,y_{j \neq \text{C}}} - \left(\frac{\partial G_M}{\partial y_{\text{Va},2}} \right)_{T,P,y_{j \neq \text{Va}}} \right] \quad (30)$$

As can be seen by the last part of Eq. (30) it is independent of the constituent in the first sublattice so it does not matter if we had chosen to use the end-members (Cr:C) and (Cr:Va) in Eq. (29).

It is not always possible to combine end-members in all possible CEF models to extract the chemical potentials of the elements, for example in the σ phase model in Section 2.4.3. But the method described below to calculate the equilibrium can also be applied for such phases.

4. Minimization with constraints

To minimize a function with constraints we apply a Lagrange equation where each equality constraint has a multiplier. When the constraint is obeyed the minimum of the Lagrange equation is the same as the original function.

4.1. Constraints

The variables in the Gibbs energy expression have several constraints. The first is that the sum of the site fractions on each sublattice is unity:

$$g_s^{\alpha} = 1 - \sum_i y_{is}^{\alpha} = 0 \quad (31)$$

For phases with ions the net charge must be zero and an external charge balance constraint must be added:

$$Q^{\alpha} = \sum_s a_s \sum_i v_i y_{is}^{\alpha} = 0 \quad (32)$$

where v_i is the charge on constituent i .

The total Gibbs energy for a system, G , is given by Eq. (1). The number of formula units of a phase α , $\aleph^{\alpha}$, must be equal to or larger than zero for a stable phase.

For a closed system we have the constraint on the amount of elements

$$f_A = \tilde{N}_A - N_A = \tilde{N}_A - \sum_{\alpha} \aleph^{\alpha} M_A^{\alpha} = 0 \quad (33)$$

where $\tilde{N}_A$ is the prescribed amount of element A.

4.2. The Lagrangian

To minimize the Gibbs energy of a system with constraints we can use a Lagrangian:

$$L = \sum_{\alpha} \aleph^{\alpha} G_M^{\alpha} + \sum_A f_A \mu_A + \sum_s \eta_s^{\alpha} g_s^{\alpha} + \sum_{\alpha} \lambda^{\alpha} Q^{\alpha} + \sum_{\psi} \gamma^{\psi} \aleph^{\psi} \quad (34)$$

where μ_A , η_s^{α} , λ^{α} are multipliers for all phases and γ^{ψ} are multipliers for all phases ψ that are unstable. From now on it will rarely be indicated which variables are kept constant at the partial derivatives, the reader is expected to understand this from the context.

4.3. The derivative of the Lagrangian with respect to the phase amount

For the partial derivative of L with respect to the amount of a stable phase α we get:

$$\frac{\partial L}{\partial \aleph^{\alpha}} = G_M^{\alpha} - \sum_A M_A^{\alpha} \mu_A = 0 \quad (35)$$

and from this equation we can understand that the Lagrange multiplier μ_A is the chemical potential of element A. For a rigorous proof see [6].

For an unstable phase ψ which is not included in the stable phase set we get:

$$\frac{\partial L}{\partial N^\psi} = G_M^\psi - \sum_A M_A^\psi \mu_A + \gamma^\psi = 0 \quad (36)$$

This means that the driving force to form a currently unstable phase, γ^ψ , can be calculated as part of the minimization and if γ^ψ becomes positive it means that the phase ψ should be added to the stable phase set. On the other hand, if the amount for a stable phase α , N^α , becomes negative it means this phase has become unstable and should be removed from the stable set. In both cases we must change the set of stable phases as discussed in section 5.6

4.4. The derivative of the Lagrangian with respect to a constituent fraction

For the partial derivative of L with respect to a constituent fraction y_{is}^z , keeping all other variables constant, we get:

$$\frac{\partial L}{\partial y_{is}^z} = N^\alpha \frac{\partial G_M^\alpha}{\partial y_{is}^z} - N^\alpha \sum_A \mu_A \frac{\partial M_A^\alpha}{\partial y_{is}^z} - \eta_s^\alpha + \lambda^\alpha \frac{\partial Q^\alpha}{\partial y_{is}^z} = 0 \quad (37)$$

We would like to use this equation in an iterative procedure to find the equilibrium and to obtain a linear correction of the difference between the current values of the constituent fractions and those of the equilibrium. We expand the partial derivative of $\frac{\partial G_M^\alpha}{\partial y_{is}^z}$ in a Taylor series of its independent variables dT , dP and dy_{jt}^z :

$$\frac{\partial G_M^\alpha}{\partial y_{is}^z} = \left(\frac{\partial G_M^\alpha}{\partial y_{is}^z} \right) + \left(\frac{\partial^2 G_M^\alpha}{\partial y_{is}^z \partial T} \right) dT + \left(\frac{\partial^2 G_M^\alpha}{\partial y_{is}^z \partial P} \right) dP + \sum_t \sum_j \left(\frac{\partial^2 G_M^\alpha}{\partial y_{is}^z \partial y_{jt}^z} \right) dy_{jt}^z \quad (38)$$

where the terms on the right hand side are calculated for the current T , P and Y and the term on the left hand side is the linearly “extrapolated” value. The last term on the right hand side is a summation over all constituents j on all sublattices t . In the rest of this paper it will often be written as just a summation over the constituents j .

Inserting Eq. (38) in Eq. (37) and changing to finite differences we get a system of linear equations depending on the corrections in ΔT , ΔP and Δy_{is}^z :

$$\sum_j \left(\frac{\partial^2 G_M^\alpha}{\partial y_{is}^z \partial y_{jt}^z} \right) \Delta y_{jt}^z - \frac{\eta_s^\alpha}{N^\alpha} + \frac{\lambda^\alpha}{N^\alpha} \frac{\partial Q^\alpha}{\partial y_{is}^z} = \sum_A \mu_A \left(\frac{\partial M_A^\alpha}{\partial y_{is}^z} \right) - \left(\frac{\partial G_M^\alpha}{\partial y_{is}^z} \right) - \left(\frac{\partial^2 G_M^\alpha}{\partial y_{is}^z \partial T} \right) \Delta T - \left(\frac{\partial^2 G_M^\alpha}{\partial y_{is}^z \partial P} \right) \Delta P \quad (39)$$

We are interested in solving this for the fraction corrections Δy_{jt}^z and can arrange the equations for $\frac{\partial L}{\partial y_{is}^z} = 0$ for the constituents i of a phase α in separate matrices. Omitting the phase and sublattice indices and for a phase with no ions and at constant P this can be written for each phase:

$$\begin{pmatrix} \frac{\partial^2 G_M}{\partial y_1^2} & \frac{\partial^2 G_M}{\partial y_1 \partial y_2} & \dots & 1 & \dots \\ \frac{\partial^2 G_M}{\partial y_1 \partial y_2} & \frac{\partial^2 G_M}{\partial y_2^2} & \dots & 1 & \dots \\ \vdots & \vdots & \ddots & \vdots & \ddots \\ 1 & 1 & \dots & 0 & \dots \\ \vdots & \vdots & \ddots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \Delta y_1 \\ \Delta y_2 \\ \vdots \\ -\frac{\eta_s}{N} \\ \vdots \end{pmatrix} = \begin{pmatrix} \sum_A \mu_A \frac{\partial M_A}{\partial y_1} - \frac{\partial G_M}{\partial y_1} - \frac{\partial^2 G_M}{\partial y_1 \partial T} \Delta T \\ \sum_A \mu_A \frac{\partial M_A}{\partial y_2} - \frac{\partial G_M}{\partial y_2} - \frac{\partial^2 G_M}{\partial y_2 \partial T} \Delta T \\ \vdots \\ 0 \\ \vdots \end{pmatrix} \quad (40)$$

The matrix on the left hand side is symmetric and the columns and rows with 1 represent the constraint that the sum of constituent fractions on each sublattice is unity. We call this the *phase matrix* and if we invert it we can express the corrections of the constituent fractions of each phase in the global potentials μ_A , ΔT and ΔP . This will be explained in detail for a specific case in Section 5.2 and generalized in Section 5.3.

5. Calculating the equilibrium

The solution is calculated by an iterative process and each iteration is divided into two steps. Before the first iteration there is an initial procedure to find a good start set of stable phases and their constitutions.

5.1. Obtaining start values by grid minimizer

An iterative procedure for calculating the equilibrium can only find a local equilibrium which depends on the initial constitution of the phases and it is thus necessary to provide a reasonable estimate of the stable phases and constitution of all phases for the first step. There are many methods for global minimization as discussed in the review by Floudas and Gounaris [26] but in thermodynamic systems there is a particular problem because some of the phases can be stable with two or more different compositions, i.e. miscibility gaps, which may not be known beforehand. There are few methods which consider this possibility and the method used here is based on the method to find the equilibrium for a system where all phases have fixed compositions. If T , P and the overall composition is known such an equilibrium is easily found by a combination of simplex and steepest decent methods and we will always find as many stable phases as we have elements and that is the maximum number possible, see Section 5.6.

The same technique can be adopted for systems with solution phases by calculating a number of gridpoints in each phase and then treat each of these as it was a separate phase. The number of gridpoints does not have to be very large even in multicomponent systems, normally a few 1000 points in each phase is sufficient even in systems with more than 10 elements and these can be calculated for each phase in parallel. After finding the gridpoints representing the equilibrium for the known composition we must check whether some of them belong to the same solution phase and, if so, check whether they can be merged into a single composition. This is not always the case as there can be miscibility gaps in solution phases. In Fig. 1 a grid generated for the phases in the Fe–Mo system at 1400 K is shown together with the convex hull. As can be seen there is a miscibility gap in the bcc phase which is metastable at this temperature.

If, for example, T is not a condition an initial search for a grid minimum is impossible. But after an iterative equilibrium calculation a value of T is determined and the grid minimizer can be used with this value of T to check if there are gridpoints below the calculated equilibrium and, if so, use these to find the global equilibrium.

If an equilibrium has already been calculated and there is only a small change in one condition, like during a calculation of a diagram, it is not necessary to perform a grid minimization every time since we can use the already calculated constitutions of the phases as start values. But it is important to check now and then to make sure that the current equilibrium is the global one as we may step into a miscibility gap that is not stable initially and may not be detected by an iterative method.

5.2. Step 1, the phase matrix

The equation for the correction of the constituent fractions will be first derived for a binary system and then generalized.

5.2.1. The phase matrix for a binary system

For a substitutional binary phase in an A–B system we will use 1 and 2 as constituent indices to emphasize that they may be different from the components of the system. The system of equations from Eq. (39) applied to this case, omitting phase and sublattice indices and for constant P is:

$$\begin{pmatrix} \frac{\partial^2 G_M}{\partial y_1^2} & \frac{\partial^2 G_M}{\partial y_1 \partial y_2} & 1 \\ \frac{\partial^2 G_M}{\partial y_1 \partial y_2} & \frac{\partial^2 G_M}{\partial y_2^2} & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \Delta y_1 \\ \Delta y_2 \\ -\frac{\eta}{N} \end{pmatrix} = \begin{pmatrix} -\frac{\partial G_M}{\partial y_1} - \frac{\partial^2 G_M}{\partial y_1 \partial T} \Delta T + \mu_A \frac{\partial M_A}{\partial y_1} + \mu_B \frac{\partial M_B}{\partial y_1} \\ -\frac{\partial G_M}{\partial y_2} - \frac{\partial^2 G_M}{\partial y_2 \partial T} \Delta T + \mu_A \frac{\partial M_A}{\partial y_2} + \mu_B \frac{\partial M_B}{\partial y_2} \\ 0 \end{pmatrix} \quad (41)$$

On the left hand side we have the phase matrix including the constraint that the sum of constituent fractions is unity.

We cannot solve Eq. (41) now as ΔT , μ_A and μ_B are not known but for reasons that will be clear later we will invert the phase matrix as we can use that several times below. We write the inverted matrix as:

$$\begin{pmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{pmatrix} = \left(\frac{\partial^2 G_M}{\partial y_i \partial y_j} \right)^{-1}$$

Inverting a matrix is not a recommended efficient mathematical procedure but in this case by inverting each phase matrix separately has advantages. The equations for the overall equilibrium become much simpler and a much larger system of equations for all stable phases with both global variables like chemical potentials and phase specific constitutions does not have to be solved. We can gain speed as the phase matrices can be calculated and inverted in parallel.

Only a part of the phase matrix is important because we are not interested in the η multipliers. We can write the solution for Δy_1 and Δy_2 as:

$$\begin{pmatrix} \Delta y_1 \\ \Delta y_2 \end{pmatrix} = \begin{pmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{pmatrix} \begin{pmatrix} -\frac{\partial G_M}{\partial y_1} - \frac{\partial^2 G_M}{\partial y_1 \partial T} \Delta T + \mu_A \frac{\partial M_A}{\partial y_1} + \mu_B \frac{\partial M_B}{\partial y_1} \\ -\frac{\partial G_M}{\partial y_2} - \frac{\partial^2 G_M}{\partial y_2 \partial T} \Delta T + \mu_A \frac{\partial M_A}{\partial y_2} + \mu_B \frac{\partial M_B}{\partial y_2} \end{pmatrix} \quad (42)$$

5.3. The general equation for the correction of constituent fractions

A system of equations like Eq. (42) can be formulated for each phase independent of its model and we can generalize an expres-

sion for Δy_{is}^z for any phase α as a function of the potentials of the system:

$$\Delta y_{is}^z = c_{iG}^z + c_{iT}^z \Delta T + c_{iP}^z \Delta P + \sum_A c_{iA}^z \mu_A \quad (43)$$

where the coefficients in this equation depend on the first and second derivatives of G_M^z of the phase:

$$\begin{aligned} c_{iG}^z &= -\sum_j e_{ij}^z \frac{\partial G_M^z}{\partial y_j^z} \\ c_{iT}^z &= -\sum_j e_{ij}^z \frac{\partial^2 G_M^z}{\partial T \partial y_j^z} \\ c_{iP}^z &= -\sum_j e_{ij}^z \frac{\partial^2 G_M^z}{\partial P \partial y_j^z} \\ c_{iA}^z &= \sum_j e_{ij}^z \frac{\partial M_A^z}{\partial y_j^z} \end{aligned} \quad (44)$$

where i is the constituent in a sublattice s and the summation over j is for all constituents in all sublattices. The summation over A is for all elements.

Eq. (43) gives a very important relation for the finite difference of a site fraction expressed as a function of the potentials T , P and μ_A , and it will be used several times below in Sections 5.5 and 6.1.

5.4. Charge balance

If some constituents have a charge we must add the differential of Eq. (32) to ensure that the phase is electrically neutral.

$$Q^z = \sum_s a_s \sum_i v_i y_{is}^z = 0 \quad (45)$$

$$\Delta Q^z = \sum_s a_s \sum_i \frac{\partial Q^z}{\partial y_{is}^z} \Delta y_{is}^z \quad (46)$$

where v_i is the charge on constituent i and

$$\frac{\partial Q^z}{\partial y_{is}^z} = a_s v_i \quad (47)$$

A phase with ions has this equation as the last row and column in its phase matrix. After inverting the phase matrix the correction of the constituent fractions, Eq. (43), will have an additional term:

$$\Delta y_{is}^z = c_{iG}^z + c_{iT}^z \Delta T + c_{iP}^z \Delta P + \sum_A c_{iA}^z \mu_A - e_{iq}^z Q^z \quad (48)$$

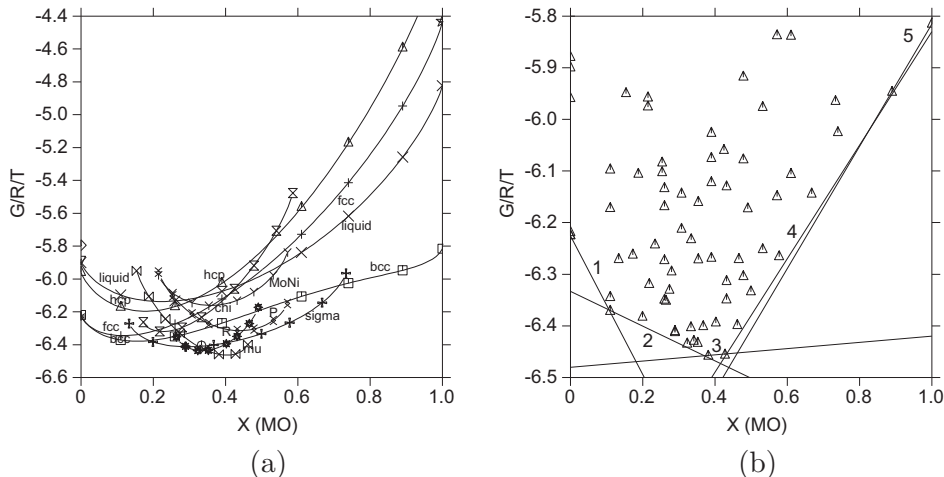


Fig. 1. In (a) the Gibbs energy curves for the phases in the Fe–Mo system have been calculated at 1400 K. Along the lines a set of gridpoints have been selected as shown. In (b) these gridpoints are shown together with the pairs of gridpoints that represent the convex hull at different compositions. We can deduce that the pair of gridpoints are sometimes in the same solution phase, sometimes in different ones. In a ternary system there will be 3 points, in a quaternary 4, etc., in each case defining a hyperplane of the lowest chemical potentials of the elements for a known composition.

where e_{iq}^z is the last column in the inverted phase matrix and Q^z is the current net charge of the phase.

5.5. Step 2, the equilibrium matrix and external conditions

After calculating the inverted phase matrices for all phases and saving them we formulate the equations to calculate new values of the potentials and phase amounts making use of the external conditions. The simplest set of external conditions is fixed T , P and amount of the elements, i.e. a closed system. For each stable phase α we will have an equation:

$$G_M^\alpha = \sum_A M_A^\alpha \mu_A \quad (49)$$

For equilibrium calculations with variable T and P we must take into account any changes in these:

$$G_M^\alpha = \sum_A M_A^\alpha \mu_A - \frac{\partial G_M}{\partial T} \Delta T - \frac{\partial G_M}{\partial P} \Delta P \quad (50)$$

Eqs. (49) or (50) ensures that the Gibbs energies of the stable phases are on a common hyperplane. The other equations depend on the external conditions as well as the amount of the stable phases.

5.5.1. Condition on the amount of the elements

The amount of each element summed over all phases is:

$$N_A - \tilde{N}_A = \sum_\alpha N^\alpha M_A^\alpha - \tilde{N}_A = 0 \quad (51)$$

where $\tilde{N}_A$ is the prescribed amount of moles of element A. The differential of N is:

$$\Delta N_A = \sum_\alpha \Delta N^\alpha M_A^\alpha + \sum_\alpha \Delta N^\alpha M_A^\alpha = 0 \quad (52)$$

From Eq. (10) we have:

$$\Delta M_A^\alpha = \sum_i \frac{\partial M_A^\alpha}{\partial y_i^z} \Delta y_i^z \quad (53)$$

where the summation over i is for all constituents so we have omitted the sublattice index. We can approximate the differentials with finite differences and for Δy_i^z we now use Eq. (43) and can write, omitting the phase superscripts and using the coefficients defined in Eq. (44):

$$\Delta M_A = \sum_i \frac{\partial M_A}{\partial y_i} \sum_B c_{iB} \mu_B - \sum_i \frac{\partial M_A}{\partial y_i} c_{iG} - \Delta T \sum_i \frac{\partial M_A}{\partial y_i} c_{iT} - \Delta P \sum_i \frac{\partial M_A}{\partial y_i} c_{iP} \quad (54)$$

where the sum over B is for all elements. For fixed T and P this becomes:

$$\Delta M_A = \sum_i \frac{\partial M_A}{\partial y_i} \sum_B c_{iB} \mu_B - \sum_i \frac{\partial M_A}{\partial y_i} c_{iG} \quad (55)$$

5.5.2. Example: a binary system with a single stable phase

We apply this to a binary A–B system with just one stable phase. The sum of the fractions of A and B in this phase must fulfill the mass balance for each element and we can insert Eq. (55) in Eq. (52):

$$\Delta N_A = N \left(\sum_i \frac{\partial M_A}{\partial y_i} \sum_B c_{iB} \mu_B - \sum_i \frac{\partial M_A}{\partial y_i} c_{iG} \right) + \Delta N M_A = 0 \quad (56)$$

and rearranging the terms we have for each element:

$$N \sum_i \frac{\partial M_A}{\partial y_i} \sum_B c_{iB} \mu_B + \Delta N M_A = N \sum_i \frac{\partial M_A}{\partial y_i} c_{iG} \quad (57)$$

Again, the sum over i should be for all constituents in all sublattices. We can now combine this with Eq. (49) to a system of linear equations:

$$\begin{pmatrix} M_A & M_B & 0 \\ N \sum_i \frac{\partial M_A}{\partial y_i} c_{iA} & N \sum_i \frac{\partial M_A}{\partial y_i} c_{iB} & M_A \\ N \sum_i \frac{\partial M_B}{\partial y_i} c_{iA} & N \sum_i \frac{\partial M_B}{\partial y_i} c_{iB} & M_B \end{pmatrix} \begin{pmatrix} \mu_A \\ \mu_B \\ \Delta N \end{pmatrix} = \begin{pmatrix} G_M \\ N \sum_i \frac{\partial M_A}{\partial y_i} c_{iG} \\ N \sum_i \frac{\partial M_B}{\partial y_i} c_{iG} \end{pmatrix} \quad (58)$$

All the terms in Eq. (58) except μ_A , μ_B and ΔN are known and this means we can calculate their new values. These can be inserted in Eq. (43) and obtain new constitutions of all phases and calculate new phase matrices and new terms in Eq. (58). Solving this we obtain new values of the potentials and we can continue to iterate until all the changes are sufficiently small, i.e. we have found the equilibrium. Note that we have a variable ΔN even if there is a single stable phase because this phase may contain molecules or vacancies and thus the number of formula units may vary with the constitution. The matrix in Eq. (58) will be called the *equilibrium matrix*.

If we have several phases and N^α becomes negative for a phase α , it should be set as unstable. The unstable phases will also have their constitutions updated at each iteration using Eq. (43) which is valid for all phases and this makes it possible to know at each iteration if an unstable phase should become stable. Care must be taken when changing the set of stable phases as described in Section 5.6.

5.5.3. Example: a binary system with two stable phases

For a binary system with fixed T and P and two stable phases α and β the system equations are (note that φ is used as phase summation index):

$$\begin{pmatrix} M_A^\alpha & M_B^\alpha & 0 & 0 \\ M_A^\beta & M_B^\beta & 0 & 0 \\ \sum_\varphi N^\varphi \sum_i \frac{\partial M_A^\varphi}{\partial y_i^z} c_{iA}^\varphi & \sum_\varphi N^\varphi \sum_i \frac{\partial M_A^\varphi}{\partial y_i^z} c_{iB}^\varphi & M_A^\alpha & M_A^\beta \\ \sum_\varphi N^\varphi \sum_i \frac{\partial M_B^\varphi}{\partial y_i^z} c_{iA}^\varphi & \sum_\varphi N^\varphi \sum_i \frac{\partial M_B^\varphi}{\partial y_i^z} c_{iB}^\varphi & M_B^\alpha & M_B^\beta \end{pmatrix} \begin{pmatrix} \mu_A \\ \mu_B \\ \Delta N^\alpha \\ \Delta N^\beta \end{pmatrix} = \begin{pmatrix} G_M^\alpha \\ G_M^\beta \\ \sum_\varphi N^\varphi \sum_i \frac{\partial M_A^\varphi}{\partial y_i^z} c_{iG}^\varphi \\ \sum_\varphi N^\varphi \sum_i \frac{\partial M_B^\varphi}{\partial y_i^z} c_{iG}^\varphi \end{pmatrix} \quad (59)$$

Iteratively solving this system for μ_A , μ_B , ΔN^α and ΔN^β together with Eq. (43) to correct the phase constitutions, updating the inverted phase matrices to obtain the new values in Eq. (59) and taking into account that the set of stable phases may change, will eventually lead to the equilibrium when the corrections of all variables become smaller than a prescribed value.

For a closed system with fixed T and P there must always be a stable equilibrium even if we may have to vary the set of stable phases to find it. But it is possible to prescribe conditions that have no solution which could happen in the next example.

5.5.4. Example: a binary system with unknown T and one stable phase prescribed

It is possible to prescribe that a phase should be stable and such a condition can either be set explicitly by the user or set automatically when calculating a phase diagram. All lines in a phase diagram represent values of the axis variables where the amount of a specific phase is zero. So when mapping a phase diagram with two axis, the condition for one of the axis variables is replaced by the condition that a phase should be stable with zero amount.

We assume that the other conditions are fixed amounts of the elements A and B and fixed P . We must have one phase stable with variable amount so we have two phases stable, one with zero amount. Note that with this set of conditions it is possible that no equilibrium exists.

For a binary system with unknown T and two stable phases, one of which, β , is prescribed to be stable with amount zero, will have an equilibrium matrix as:

$$\begin{pmatrix} M_A^\alpha & M_B^\alpha & 0 & -\frac{\partial G_M^\alpha}{\partial T} \\ M_A^\beta & M_B^\beta & 0 & -\frac{\partial G_M^\beta}{\partial T} \\ N^\alpha \sum_i \frac{\partial M_i^\alpha}{\partial y_i^\alpha} C_{iA}^\alpha & N^\alpha \sum_i \frac{\partial M_i^\alpha}{\partial y_i^\alpha} C_{iB}^\alpha & M_A^\alpha & -N^\alpha \sum_i \frac{\partial M_i^\alpha}{\partial y_i^\alpha} C_{iT}^\alpha \\ N^\alpha \sum_i \frac{\partial M_i^\beta}{\partial y_i^\beta} C_{iA}^\alpha & N^\alpha \sum_i \frac{\partial M_i^\beta}{\partial y_i^\beta} C_{iB}^\alpha & M_B^\alpha & -N^\alpha \sum_i \frac{\partial M_i^\beta}{\partial y_i^\beta} C_{iT}^\alpha \end{pmatrix} \begin{pmatrix} \mu_A \\ \mu_B \\ \Delta N^\alpha \\ \Delta T \end{pmatrix} = \begin{pmatrix} G_M^\alpha \\ G_M^\beta \\ N^\alpha \sum_i \frac{\partial M_i^\alpha}{\partial y_i^\alpha} C_{iG}^\alpha \\ N^\alpha \sum_i \frac{\partial M_i^\beta}{\partial y_i^\beta} C_{iG}^\alpha \end{pmatrix} \quad (60)$$

In this system of equations there is only one phase with variable amount, N^α , as $\Delta N^\beta = 0$. Note that we take into account that the Gibbs energy of the phases varies with ΔT according to Eq. (50). When we calculate the correction of the constituent fractions, Δy_i , we must take into account they depend on ΔT according to Eq. (43).

5.5.5. Example: a binary system with a condition of a chemical potential

The final example is for a binary system with fixed T , P , the total amount of one elements, N_A , and the chemical potential, μ_B of the other. With two phases stable we have:

$$\begin{pmatrix} M_A^\alpha & 0 & 0 \\ M_A^\beta & 0 & 0 \\ \sum_\varphi N^\varphi \sum_i \frac{\partial M_i^\varphi}{\partial y_i^\varphi} C_{iA}^\varphi & M_A^\alpha & M_A^\beta \end{pmatrix} \begin{pmatrix} \mu_A \\ \Delta N^\alpha \\ \Delta N^\beta \end{pmatrix} = \begin{pmatrix} G_M^\alpha - M_B^\alpha \mu_B \\ G_M^\beta - M_B^\beta \mu_B \\ \sum_\varphi N^\varphi \sum_i \frac{\partial M_i^\varphi}{\partial y_i^\varphi} (C_{iG}^\varphi - C_{iB}^\varphi \mu_B) \end{pmatrix} \quad (61)$$

The summation over φ is for the stable phases.

In most examples above T and P have been fixed and in one example a chemical potential is fixed. We cannot have all conditions as potentials because then we are trying to minimize the Gibbs–Duhem relation which is always zero. At least one condition must be an extensive or normalized property.

5.6. Changing the set of stable phases

When the amount of a stable phase becomes negative during an iteration it means this phase should be removed from the set of stable phases. And if the driving force for an unstable phase becomes positive that phase should be added. The driving force, γ^ψ for an unstable phase ψ is calculated by Eq. (36):

$$\gamma^\psi = \sum_A \mu_A M_A^{\psi_i} - G_M^\psi \quad (62)$$

where the sum over A is for all elements. If T and P are variable the ΔT and ΔP are also included in this equation as in Eq. (50).

It is basically a trivial operation to add or remove a phase but we have to be careful that the same phase is not removed and added during every other iteration. Normally we should allow a few iterations after a change in the set of stable phases before another change is allowed.

It can happen that the amount of the only stable phase becomes negative and that is clearly an impossible situation. It may indicate that the set of external conditions is unreasonable or the thermodynamic models are bad or the implementation of the algorithm is wrong.

A third case that may cause problem is when more phases become stable than allowed by the Gibbs phase rule:

$$f = n - p + 2 \quad (63)$$

where f is the degrees of freedom, n number of elements, p number of stable phases and 2 represents variable T and P . In order to calculate an equilibrium we must have set so many conditions that f is zero. For a binary system that means 4 conditions. If one condition is constant P , we have at most have 3 phases stable.

For a calculation in a binary system with fixed T and P we cannot have more than 2 stable phases. If a third phase wants to become stable this phase must replace one of the already stable phases. If this happens while calculating a binary phase diagram the software may change the set of conditions and add a variable T and set the new phase as fixed with zero amount to find the invariant three phase equilibrium. This will be discussed in a later paper on phase diagram calculations.

5.7. Normalized state variables as conditions

In the examples above the total amount of the elements has been fixed. This simplifies the equations because if we use normalized state variables like mole fractions rather than total amounts we must take into account the derivatives of the normalized property when deriving the equations for the system and that these may change during the iterations.

Conditions on extensive properties can in general be set on a global variable (summed over all stable phases) or for a specific phase and it can be a total value or a normalized one (normalized per moles, mass or volume). For a global total property we can formulate the condition like for N_A in Eq. (51).

For a normalized phase specific variable we have, using Z for the variable and K for the normalized quantity:

$$z^\alpha = \frac{Z_M^\alpha}{K_M^\alpha} \quad (64)$$

If the prescribed value is $\tilde{z}^\alpha$ we have

$$\Delta z^\alpha = \tilde{z}^\alpha - z^\alpha = \frac{\Delta Z_M^\alpha - z^\alpha \Delta K_M^\alpha}{K_M^\alpha} \quad (65)$$

where the subscript M means per mole formula unit of the phase. The finite difference Δz^α can be expressed in the model variables of the phase, T , P and y_j^α , as:

$$\Delta Z_M^\alpha = \frac{\partial Z_M^\alpha}{\partial T} \Delta T + \frac{\partial Z_M^\alpha}{\partial P} \Delta P + \sum_j \frac{\partial Z_M^\alpha}{\partial y_j^\alpha} \Delta y_j^\alpha \quad (66)$$

and similarly for ΔK_M^α . Here we can again use Eq. (43) to express Δy_j^α as a function of ΔT , ΔP and the chemical potentials μ_A .

To prescribe a global value for the whole system we can simply sum the normalized variable and the normalizing factor for all stable phases:

$$Z = \sum_\alpha N^\alpha Z_M^\alpha \quad (67)$$

$$K = \sum_\alpha N^\alpha K_M^\alpha \quad (68)$$

and for a global property $z = \frac{Z}{K}$ we can express the deviation from the prescribed value $\tilde{z}$ using finite differences as:

$$\Delta z = \tilde{z} - z = \frac{1}{K} \left[\sum_\alpha N^\alpha (\Delta Z_M^\alpha - z \Delta K_M^\alpha) + \sum_\alpha (Z_M^\alpha - z K_M^\alpha) \Delta N^\alpha \right] \quad (69)$$

As before we can calculate ΔZ_M^α and ΔK_M^α as functions of ΔT , ΔP and the chemical potentials μ_A using Eqs. (66) and (43).

A typical use of Eq. (69) is when there is a condition on the mole or mass fraction of an element. More details can be found in the full documentation [11].

5.8. Generalizing the equilibrium matrix

All examples shown above have been for binary cases. The way to calculate corrections to the constituent fractions, Eq. (43), is completely general and can be used for all phases in a multicomponent system. The OC software has been prepared to calculate these in parallel.

As shown with the examples the equilibrium equations must be adopted to the conditions set by the user and they will also change if the number of stable phases changes during the iterations. The coefficients calculated from the inverted phase matrix, Eq. (44), are necessary to construct the equations for the equilibrium matrix. The method presented here is valid for systems with any number of components.

It is possible to combine solving the phase matrix and the equilibrium matrix but that may generate a huge matrix if all phases are taken into account. If only the stable phases are considered, like in the implementation by Lukas et al. [8], there is a problem to know when to change the set of stable phases. In neither case one can use parallelization to speed up the time-consuming calculations of all values of the Gibbs energies and their derivatives from the models.

6. Calculating properties using the results from an equilibrium calculation

Most of the thermodynamic properties like G , μ_A , S , V , H , T , etc. are directly available after an equilibrium calculation. But some properties, like the heat capacity, C_p require knowledge how the system will react to a change in T , for example if the constitution of a gas phase changes. The enthalpy of a system is related to the Gibbs energy and entropy as:

$$H = G + TS \quad (70)$$

and the heat capacity at constant P is defined as:

$$C_p = \left(\frac{\partial H}{\partial T} \right)_{P, N_A} \quad (71)$$

and this should be calculated for constant overall composition, which means we must consider the change in all internal degrees of freedom in order to get the correct value. In the Thermo-Calc software [27] this has been implemented as a “dot derivative” which allows us to calculate a partial derivative of a state variable with respect to any other state variable, for example “ $G.T$ ” or “ $H.T$ ”, for the current set of conditions. Such a calculation require that the state variable before the “dot” must not be a condition and the variable after the dot must be a condition in the last equilibrium calculation.

The value of $G.T$ will normally be $-S$ of course but if we have used a fixed chemical potential or a fixed phase as a condition the value will be different because S is defined as the derivative $-\left(\frac{\partial G}{\partial T}\right)_{P, N_A}$ for a closed system at constant P .

The dot derivative facility is necessary for calculating the heat capacity, $H.T$, and also very useful for calculating many other properties for example the slope of the lines in a phase diagram as “ $x(\text{liquid}, A).T$ ”. This notation means the change of the mole fraction of element A in the liquid phase with T . The dot derivative method provides such values without resorting to numerical derivation.

Many other uses can be made of the dot derivative but the user must be aware that “ $G.X(A)$ ” is not the chemical potential of A , as

explained in Section 3.2. The chemical potential is a separate state variable, which is obtained by the symbol “ $MU(A)$ ”. If T and P are constant and all other conditions are of the type $N_A = \text{value}$, then “ $G.N(A)$ ” is the same as $MU(A)$ according to Eq. (20). We must always pay special attention which variables are kept constant when dealing with partial derivatives.

The dot derivative facility was developed and implemented in the Thermo-Calc software by Bo Jansson as part of the POLY module [7] but this particular part of the software was never documented. This attempt to implement and documentate the same feature in OC is to honor his memory.

6.1. Calculation of $H.T$

This calculation is made by combining the values of appropriate coefficients for Eq. (44) so that the derivative can be obtained without a new equilibrium calculation for a small increment in T . Note that the value calculated by $H.T$ is equal to C_p only if the conditions used for the last calculation are for a closed system at constant T and P .

If the system has several stable phases each of them contributes to the heat capacity as well as the change of the phase amounts. This means the latent heat is included in the $H.T$ calculation. If there are internal degrees of freedom in a phase, like the fraction of species in a gas that may vary with T , this will give a contribution to the configurational entropy and thus to the heat capacity. The enthalpy can be written as a sum over the stable phases:

$$H = \sum_{\alpha} N^{\alpha} H_M^{\alpha} \quad (72)$$

and the change ΔH for a small change ΔT can be written

$$\Delta H = \sum_{\alpha} \Delta N^{\alpha} H_M^{\alpha} + \sum_{\alpha} N^{\alpha} \left[\left(\frac{\partial H_M^{\alpha}}{\partial T} \right)_{P, y_i} \Delta T + \sum_i \left(\frac{\partial H_M^{\alpha}}{\partial y_i} \right)_{P, y_{j \neq i}} \Delta y_i \right] \quad (73)$$

where all terms except ΔN^{α} and Δy_i can be obtained from the results of the most recent equilibrium calculation. Considering a small change ΔT , we can add this as an additional variable in the equilibrium matrix used to calculate the equilibrium. For this reason T must be a condition because otherwise we cannot add a derivative with respect to T .

In the equilibrium matrix we simply add a line representing the variable ΔT and for all previous lines in the equilibrium matrix also a column representing the derivatives with respect to T . The right hand side contains only zeros except for ΔT . For the equilibrium matrix in Eq. (58) this would be:

$$\begin{pmatrix} M_A & M_B & 0 & s_{14} \\ N \sum_i \frac{\partial M_A}{\partial y_i} c_{iA} & N \sum_i \frac{\partial M_A}{\partial y_i} c_{iB} & M_A & s_{24} \\ N \sum_i \frac{\partial M_B}{\partial y_i} c_{iA} & N \sum_i \frac{\partial M_B}{\partial y_i} c_{iB} & M_B & s_{34} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \Delta \mu_A \\ \Delta \mu_B \\ \Delta N \\ \Delta T \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (74)$$

where the terms in 4th column are:

$$s_{14} = -\frac{\partial G_M}{\partial T} \quad (75)$$

$$s_{24} = -N \sum_i \frac{\partial M_A}{\partial y_i} c_{iT} \quad (76)$$

$$s_{34} = -N \sum_i \frac{\partial M_B}{\partial y_i} c_{iT} \quad (77)$$

We solve this system of equations once to obtain the values of $\Delta \mu_A$, $\Delta \mu_B$ and ΔN , the value of ΔT is unity and the values of $\Delta \mu_A$ and $\Delta \mu_B$ are related to this change. The value of Δy_i can be calculated from Eq. (43) using these terms:

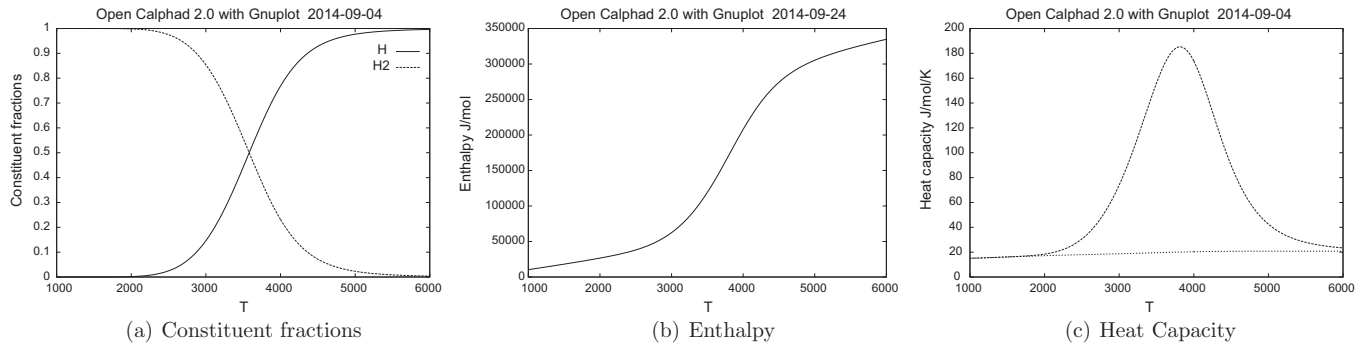


Fig. 2. Even a pure element can have a complex heat capacity curve as shown for gaseous hydrogen. In (a) the fractions of H and H_2 are shown as function of T , between 3000 and 4000 K the H_2 molecule dissociates into atomic H. In (b) the enthalpy change is shown as function of T . Note the drastic increase when the H_2 molecule dissociates. In (c) the full line is the heat capacity curve corresponding to the enthalpy curve in (b), the dashed line the heat capacity without taking the variation in constituent fractions into account.

$$\Delta y_{is} = c_{iT} \Delta T + \sum_A c_{iA} \Delta \mu_A \quad (78)$$

To demonstrate the importance of including all terms we show in Fig. 2 the calculation of the variation the constituent fractions, the enthalpy and C_p with T for a very simple system: pure hydrogen gas with the two molecules, H_2 and H. The value of $\sum_i x_i \left(\frac{\partial H_i}{\partial T} \right)_{P, y_i}$ as C_p , i.e. for fixed constitution but not fixed overall composition, and not considering any variation in the constituent fractions or phase amounts, are shown as a dashed line in Fig. 2. This is a serious error as changes in the configuration will contribute to the heat capacity when we model defects or short or long range ordering in any phase.

6.2. Calculation of $\chi(\text{liquid}, A) \cdot T$

Only the basic transcription into variables that can be calculated is given here:

$$\begin{aligned} \frac{\partial x_A^L}{\partial T} &= \frac{\partial}{\partial T} \left(\frac{M_A^L}{\sum_B M_B^L} \right) = \frac{\frac{\partial M_A^L}{\partial T} \sum_B M_B^L - M_A^L \sum_B \frac{\partial M_B^L}{\partial T}}{(\sum_B M_B^L)^2} \\ &= \frac{1}{\sum_B M_B^L} \left(\sum_i \frac{\partial M_A^L}{\partial y_i} \frac{\partial y_i}{\partial T} - x_A^L \sum_B \sum_i \frac{\partial M_B^L}{\partial y_i} \frac{\partial y_i}{\partial T} \right) \end{aligned} \quad (79)$$

where $\frac{\partial y_i}{\partial T}$ can be approximated by Δy_i for $\Delta T = 1$ and calculated in the same way as in Section 6.1. If liquid is the only stable phase and we have a closed system the result will be zero but if there are other phases stable it will be the slope of the liquidus hypersurface in the direction of element A, which is useful in simulations of solidification. It is a significant simplification to obtain this property as a “dot derivative” rather than by calculating several equilibria and use numerical derivatives.

7. Conclusions

In this paper we describe a new implementation of a well-established algorithm [6] for thermodynamic equilibrium calculations, a field where many still make mistakes calculating partial derivatives of state variables with the wrong set of conditions. The intention is that this open source software will be used and integrated in many different applications both theoretical and practical. The hope is that this paper can reduce some of the confusion and mystique about equilibrium calculations, in particular how to calculate and use the chemical potential.

The global minimization technique used here is simple but as far as we know the best one to detect unexpected miscibility gaps

in the phases. The technique of calculating “dot derivatives” has never been documented before and it is essential to get correct values for properties like the heat capacity in gases and liquids with short range order as well as in solids with defects and other internal degrees of freedom. This technique as well as the minimization as a whole relies on the implementation of analytical first and second derivatives of the Gibbs energy per mole formula unit, as provided by the General Thermodynamic Package (GTP), which is part of the OC package [11].

The OC software package has a GNU license and the source code and documentation of all parts of this software are available on the web site [11]. Collaborations in this development are invited, for new models, minimizers and other applications, and to simplify this the source code is also available as a repository called open-calphad at [12]. Implementation of property and phase diagrams calculations is ongoing and after that the most important task on the list to develop is an assessment module for thermodynamic model parameters.

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References

- [1] N. Saunders, A.P. Miodownik, Calphad, in: R.W. Cahn (Ed.), vol. 1, Pergamon Press, Oxford, 1998.
- [2] L. Kaufman, Calphad 25 (2001) 141.
- [3] H.L. Lukas, S.G. Fries, B. Sundman, *Computational Thermodynamics*, Cambridge Univ Press, 2007.
- [4] M.H.A. Piro, S. Simunovic, T.M. Besmann, B.J. Lewis, W.T. Thompson, *Comput. Mater. Sci.* 67 (2012) 266.
- [5] G. Eriksson, E. Rosen, *Chem. Scripta* 4 (1973) 193.
- [6] M. Hillert, *Physica* 103B (1981) 31.
- [7] B. Jansson, Ph.D. Thesis, Physical Metallurgy, KTH Stockholm, Sweden, 1984.
- [8] H.L. Lukas, J. Weiss, E.-Th. Henig, *Calphad* 6 (1982) 229.
- [9] K. Shobu, T. Taburu, *Mater. Trans.* 46 (2005) 1175.
- [10] B. Sundman, M. Palumbo, U. Kattner, S.G. Fries, *Integrating Materials and Manufacturing Innovation (IMMI)* (2015), <http://dx.doi.org/10.1186/s40192-014-0029-1>.
- [11] <http://www.opencalphad.org>.
- [12] The Opencalphad Repository. <<http://github.com>>.
- [13] A. van der Walle, *Calphad* 33 (2009) 266.
- [14] J.-L. Fattebert, M.E. Wickett, P.E.A. Turchi, *Acta Mater.* 62 (2014) 89.
- [15] M. Palumbo, S.G. Fries, T. Hammerschmidt, T. Abe, J.-C. Crivello, A. Breidi, J.-M. Joubert, R. Drautz, *Comput. Mater. Sci.* 81 (2014) 433.
- [16] A. Kroupa, *Comput. Mater. Sci.* 66 (2013) 3.

- [17] N. Moelans, F. Wendler, B. Nestler, *Comput. Mater. Sci.* 46 (2009) 479.
- [18] X.-G. Lu, Z. Wang, Y. Cui, Z. Jin, *Chin. Sci. Bull.* 59 (2014) 1662.
- [19] I.-H. Jung, *Calphad* 34 (2010) 332.
- [20] G.B. Olson, *Acta Mater.* 13 (2013) 771.
- [21] H.K.D.H. Bhadeshia, *Scripta Mater.* 70 (2014) 12.
- [22] B. Sundman, J. Ågren, *J. Phys. Chem. Solids* 42 (1981) 297.
- [23] M. Hillert, *J. Alloys Compd.* 320 (2001) 161.
- [24] M. Hillert, B. Jansson, B. Sundman, J. Ågren, *Metall. Trans. A* 16A (1985) 261.
- [25] J. Rogal, S.V. Divinski, M.W. Finnis, A. Glensk, J. Neugebauer, S. Schwalow, M.H.F. Sluiter, B. Sundman, *Phys. Status Solidi B* 251 (2014) 97.
- [26] C.A. Floudas, C.E. Gounaris, *J. Global Optim.* 45 (2009) 3.
- [27] J.-O. Andersson, T. Helander, L. Höglund, P. Shi, B. Sundman, *Calphad* 26 (2002) 273.